

Yiliang Wan

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EDUCATION

National University of Singapore

Ph.D. Candidate in Computer Science (Advisor: Jialin Li)

January 2024 – Present

Singapore

- Research interests: Distributed systems and operating systems.

National University of Singapore

M.Sc. in Computer Science (Advisor: Jialin Li)

August 2022 – December 2023

Singapore

- GPA: 5.0/5.0
- Thesis: *Accelerating Consensus with Network Ordering in Wireless Networks*

Beihang University

B.Eng. in Computer Science and Technology

September 2017 – June 2022

Beijing, China

- GPA: 3.83/4.00

RESEARCH OVERVIEW

Co-Designing State Machine Replication with Datacenter Networks

Co-designed a replication protocol and network runtime that exploit predictable timing in datacenter networks. Reduces pipeline stalls by decoupling round length from network propagation delay and minimizing processing jitter, achieving $O(1)$ amortized bottleneck message complexity and 3.92 million operations/s without batching on five RDMA replicas.

Operating System Support for Full-Stack Simulation

Developed OS support for cluster-scale full-stack simulation in Linux, integrating simulation-aware scheduling, memory performance control, timed communication, and cross-host coordination. Reproduced physical-system measurements within 9% across seven end-to-end metrics, with $2.6\text{--}24\times$ wall-clock slowdown.

Generative Recommendation Serving

Investigating GPU-CPU co-serving for generative recommendation models, targeting sub-100ms latency, large per-user KV caches, and fragmented interaction-driven traffic. [[Simulation platform](#)]

PUBLICATIONS AND PREPRINTS

Ossim: OS-Driven Live Simulation for Cluster-Scale Full-Stack Evaluation

Yiliang Wan, Haifeng Sun, Yihan Yang, Jonas Kaufmann, Antoine Kaufmann, Jialin Li.

The 17th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys '26), 2026. [[Paper](#)] [[Extended paper](#)]

Reaching Agreement Among Reasoning LLM Agents

Chaoyi Ruan, Yiliang Wan, Ziji Shi, Jialin Li.

arXiv preprint arXiv:2512.20184, 2025. [[Preprint](#)]

Reducing Pipeline Bubbles in Message-Efficient State Machine Replication

Yiliang Wan, Inho Choi, Nitin Shivaraman, Akshaye Shenoi, Xiang Liu, Tao Luo, Haifeng Sun, Chaoyi Ruan, Jialin Li.

arXiv preprint arXiv:2507.12792, 2025. [[Preprint](#)]

POSTERS

Practical Network Synchrony for Building Efficient Distributed Protocols

Yiliang Wan, Nitin Shivaraman, Akshaye Shenoi, Xiang Liu, Tao Luo, Jialin Li.

The 31st ACM Symposium on Operating Systems Principles (SOSP '25), Poster Session, 2025. [[Poster](#)]

EXPERIENCE

Huawei - Research Intern	July 2025 – February 2026
Markov Lab	Shanghai, China & Singapore
Datacenter Simulation and Generative Recommendation Infrastructure	
Worked on full-stack simulation and GPU-CPU co-serving for generative recommendation.	
National University of Singapore - Research Assistant	November 2022 – November 2023
NUS Systems Lab (Advisor: Jialin Li)	Singapore
Network-Ordered Consensus for Wireless Networks	
Co-designed consensus protocols with network ordering in wireless sensor networks.	
Beihang University - Research Assistant	August 2021 – June 2022
Operating System Lab (Advisor: Lei Wang)	Beijing, China
The MOS Microkernel	
Developed and optimized the inter-process communication (IPC) subsystem on AArch64.	
The Shyper Embedded Hypervisor	
Developed device drivers for an embedded type-1 hypervisor on AArch64.	

TEACHING

Distributed Systems (CS5223)	
National University of Singapore	
Teaching Assistant	Spring 2026
Teaching Assistant	Spring 2025
Teaching Assistant	Spring 2024
Introduction to Operating Systems (CS2106)	
National University of Singapore	
Teaching Assistant	Fall 2024

AWARDS

NUS Research Scholarship	2024
National University of Singapore	
Academic Excellence Award	2023
National University of Singapore	
Outstanding Student Award	2021
Beihang University	